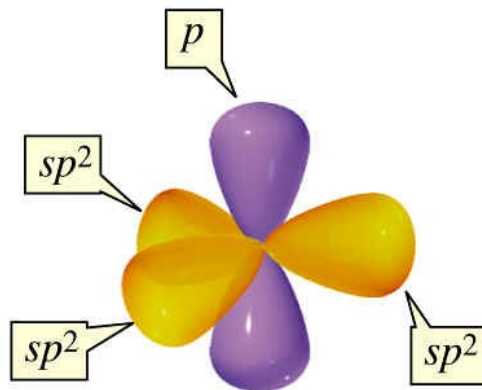
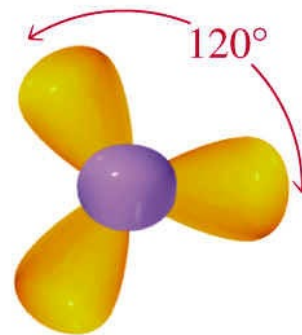


sp^2 hybridization



side view



top view

C atom: $1s^2$, $2s^2$, $2p^2$
 ignore

$$\psi_{nlm} = R_{nl}(r) Y_{lm}(\theta, \phi)$$

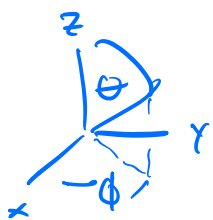
s orbitals

$$Y_{00} \sim \text{const.}$$

$$Y_{10} \sim \cos \theta$$

$$Y_{11} \sim \sin \theta e^{i\phi}$$

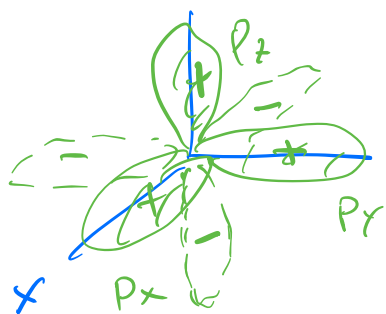
$$Y_{1-1} \sim \sin \theta e^{-i\phi}$$



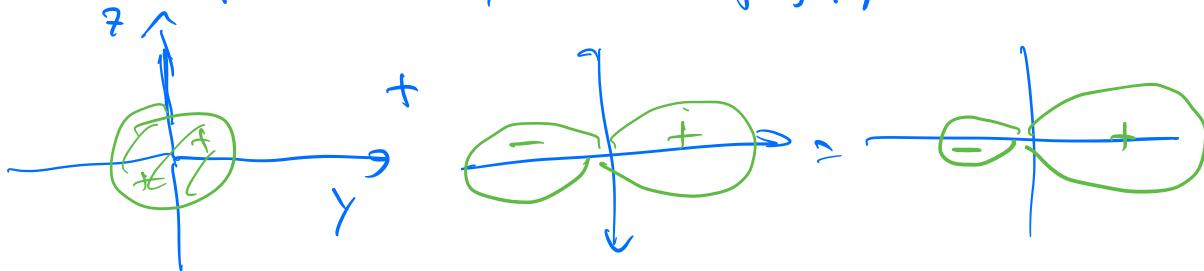
$$|p_x\rangle \sim Y_{11} - Y_{1-1} \sim \sin \theta \cos \phi$$

$$|p_y\rangle \sim Y_{11} + Y_{1-1} \sim \sin \theta \sin \phi$$

$$|p_z\rangle \sim Y_{10} \sim \cos \theta$$



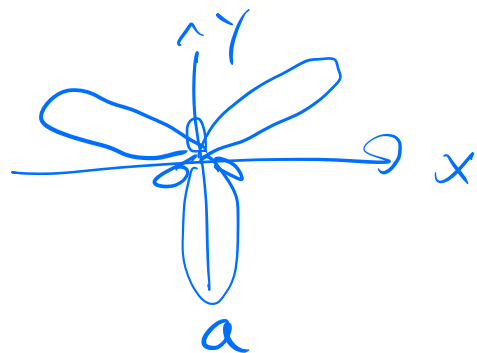
Step 2: hybridize p_x, p_y with s



$$sp^2_a = \frac{1}{\sqrt{3}} |2s\rangle - \sqrt{\frac{2}{3}} |2p_x\rangle$$

$$sp^2_b = \frac{1}{\sqrt{3}} |2s\rangle + \sqrt{\frac{2}{3}} \left\{ \frac{\sqrt{3}}{2} |2p_x\rangle + \frac{1}{2} |2p_y\rangle \right\}$$

$$sp^2_c = \frac{1}{\sqrt{3}} |2s\rangle + \sqrt{\frac{2}{3}} \left\{ -\frac{\sqrt{3}}{2} |2p_x\rangle + \frac{1}{2} |2p_y\rangle \right\}$$

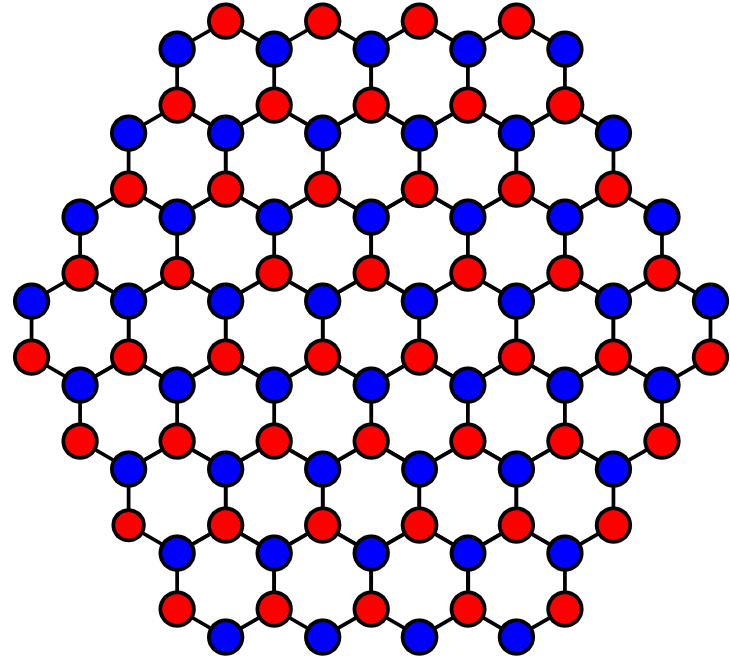
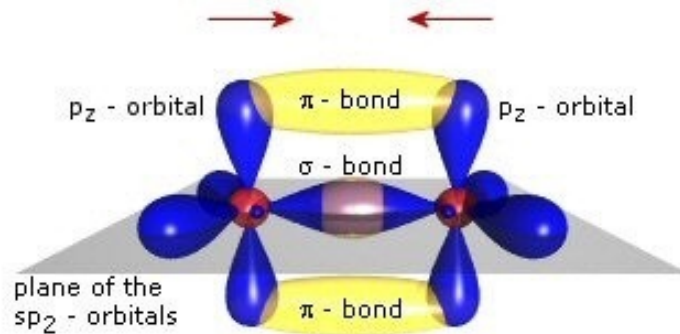


in graphene: sp^2_{abc} form σ -bonds
(strong bonding)

$|p_z\rangle$ orbitals will
form our conduction
bands

(π -bands)

Graphene electronic bandstructure



Summary of diatomic molecule

① starting orbitals $\psi_1(r), \psi_2(r)$ $|\psi_1\rangle, |\psi_2\rangle$

guess: $|\psi\rangle = u_1 |\psi_1\rangle + u_2 |\psi_2\rangle$

② tight binding model, minimize $\langle E \rangle$
(maximize)

$$\langle E \rangle = \langle \psi | H | \psi \rangle$$

$$\text{using } \frac{d\langle E \rangle}{du_{1,2}^*} = 0 \quad (\text{set } E_0 = 0)$$

$$\textcircled{3} \quad \begin{pmatrix} 0 & H_{12} \\ H_{12} & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = E \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad \begin{matrix} E = \pm H_{12} \\ \psi_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{matrix}$$

Can diagonalize using $U = (\psi_1, \psi_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$H' = U^\dagger H U \quad \text{"}^\dagger\text{"} = \text{transpose complex conjugate}$$

$$H' = \frac{H}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \frac{H}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{H}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} H & 0 \\ 0 & -H \end{pmatrix}$$

$$U^\dagger |\psi_+\rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$U^\dagger |\psi_-\rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Electronic Bandstructure of Graphene – tight binding model

- ① Start with Bloch functions 'living' on A and B sublattices

$$\phi_A = \frac{1}{\sqrt{N}} \sum_{\vec{R}_A} e^{i\vec{k} \cdot \vec{R}_A} \psi(\vec{r} - \vec{R}_A)$$

$\leftarrow p_z$ orbital on A atom
position of A atom

$$\phi_B = \frac{1}{\sqrt{N}} \sum_{\vec{R}_B} e^{i\vec{k} \cdot \vec{R}_B} \psi(\vec{r} - \vec{R}_B)$$

$$\psi_{\vec{k}} = C_A \phi_A + C_B \phi_B$$

- ② minimize energy as we did for atomic molecule

$$\begin{pmatrix} 0 & H_{AB} \\ H_{BA} & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = E \begin{pmatrix} C_A \\ C_B \end{pmatrix}$$

$E = 0$ for E_{2p_z}

Electronic Bandstructure of Graphene – tight binding model

solution is same as for the
diatomic molecule $E = \pm H_{AB}$

but... H_{AB} depends on $\vec{k}$

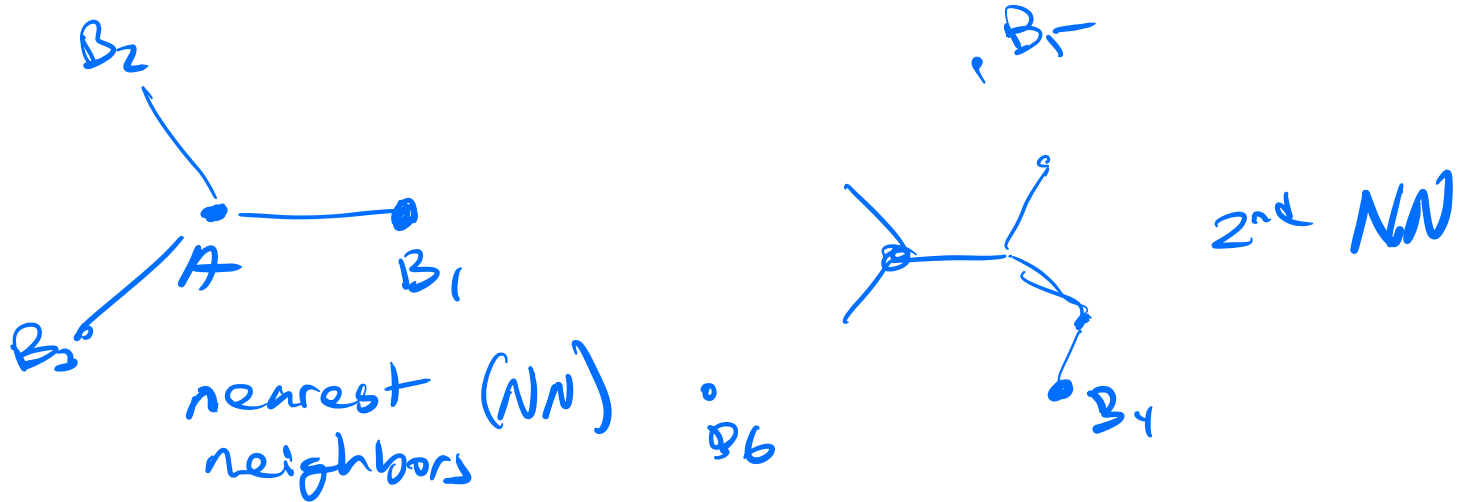
$$H_{AB} = \langle \Phi_A | \mathcal{H} | \Phi_B \rangle$$

$$= \frac{1}{N} \int \sum_{\vec{R}_A} e^{-i\vec{k} \cdot \vec{R}_A} \Phi^*(\vec{r} - \vec{R}_A) \mathcal{H} \sum_{\vec{R}_B} e^{i\vec{k} \cdot \vec{R}_B} \Phi(\vec{r} - \vec{R}_B) d\vec{r}$$

for each $\vec{R}_A$:

$$e^{-i\vec{k} \cdot \vec{R}_A} \cdot \sum_{\vec{R}_B} e^{i\vec{k} \cdot \vec{R}_B} \cdot \int \Phi^*(\vec{r} - \vec{R}_A) \mathcal{H} \Phi(\vec{r} - \vec{R}_B) d\vec{r}$$

Electronic Bandstructure of Graphene – tight binding model



Assume that integral $\neq 0$ except for nearest neighbors

for NN,
$$\int \psi^*(\vec{r}-\vec{R}_n) H \psi(\vec{r}-\vec{R}_0) d\vec{r} = t$$

 (negative)
 $t \approx -3.033 \text{ eV}$

Electronic Bandstructure of Graphene – tight binding model

for each R_A :

$$e^{-i\vec{k} \cdot \vec{R}_A} \sum_{NN} e^{i\vec{k} \cdot \vec{R}_0} \cdot t = t \cdot \sum_{NN} e^{i\vec{k} \cdot (\vec{R}_0 - \vec{R}_A)} \vec{s}$$

sum over A atoms $\rightarrow$ factor of N

$$H_{AB} = t \cdot S(k)$$
$$S(k) = \sum_{NN} e^{i\vec{k} \cdot \vec{s}}$$

$$H_{BA} = t S^*(k) \quad (\vec{s} \rightarrow -\vec{s})$$

$$\begin{pmatrix} 0 & tS(k) \\ tS^*(k) & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = E(k) \begin{pmatrix} C_A \\ C_B \end{pmatrix}$$

$$H - \varepsilon I \begin{pmatrix} -\varepsilon & tS(k) \\ tS^*(k) & -\varepsilon \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\det(H - \varepsilon I) = 0$$

$$\varepsilon^2 - t^2 S^* S = 0$$

$$\varepsilon = \pm |t| \sqrt{S^* S}$$

Electronic Bandstructure of Graphene – tight binding model

$$\varepsilon(\vec{k}) = \pm |t| \sqrt{|S_k|^2} = \pm |t| \sqrt{S_k^* S_k}$$

Evaluating $S(\vec{k}) = \sum_i e^{i\vec{k} \cdot \vec{S}_i} \quad i=1,2,3$

method 1: 'brute force'

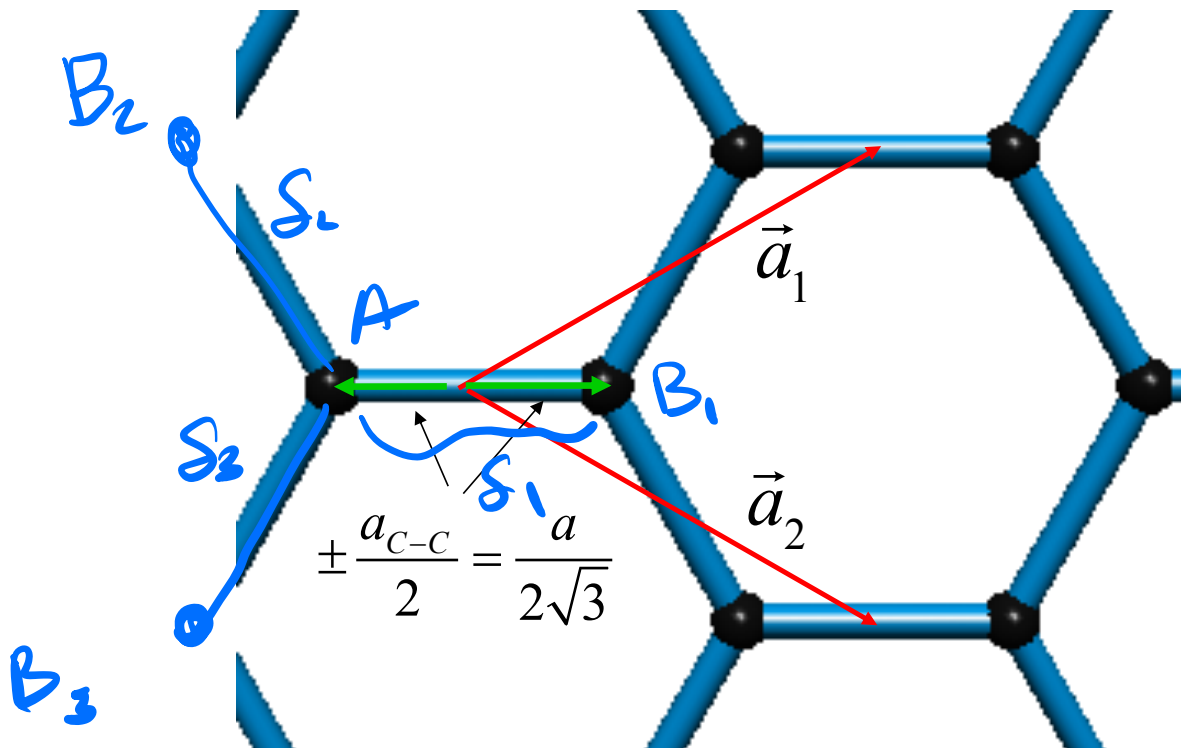
$$\vec{S}_1 = a\hat{i} \quad \vec{S}_2 = a\left(-\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right) \quad \vec{S}_3 = a\left(-\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$$

* here $a = a_{CC}$

$$\vec{k} = k_x\hat{i} + k_y\hat{j}$$

$$\begin{aligned} S(\vec{k}) &= e^{ik_x a} + e^{-ik_x \frac{a}{2}} e^{ik_y \frac{a\sqrt{3}}{2}} + e^{ik_x \frac{a}{2}} e^{-ik_y \frac{a\sqrt{3}}{2}} \\ &= e^{ik_x a} + e^{-ik_x \frac{a}{2}} \cdot 2 \cos\left(\frac{\sqrt{3}}{2} k_y a\right) \end{aligned}$$

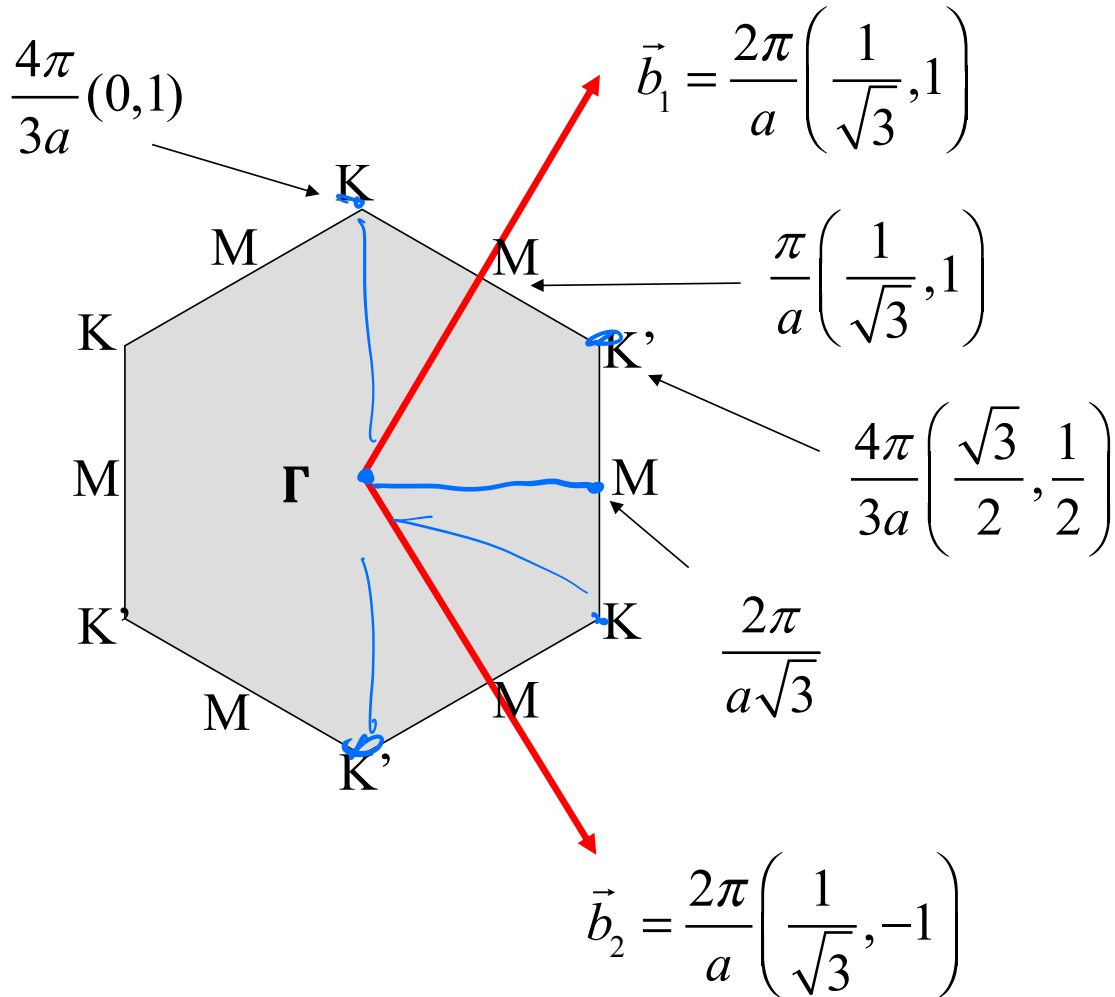
$$\begin{aligned}
|S| &= \sqrt{S^* S} \\
&= \left[e^{-ik_x a} + e^{ik_x \frac{a}{2}} \cdot 2\cos\left(\frac{\sqrt{3}}{2} k_y a\right) \right] \left[e^{ik_x a} + e^{-ik_x \frac{a}{2}} \cdot 2\cos\left(\frac{\sqrt{3}}{2} k_y a\right) \right] \\
&= 1 + \underbrace{\left(e^{i \cdot \frac{3}{2} k_x a} + e^{-i \cdot \frac{3}{2} k_x a} \right)}_{2\cos\left(\frac{3}{2} k_x a\right)} \cdot 2\cos\left(\frac{\sqrt{3}}{2} k_y a\right) + \underbrace{4\cos^2\left(\frac{\sqrt{3}}{2} k_y a\right)}_{2+2\cos(\sqrt{3} k_y a)} \\
&= 3 + 2\cos(\sqrt{3} k_y a) + 4\cos\left(\frac{\sqrt{3}}{2} k_y a\right)\cos\left(\frac{3}{2} k_x a\right) \\
E(k) &= \pm t \sqrt{3 + 2\cos(\sqrt{3} k_y a) + 4\cos\left(\frac{\sqrt{3}}{2} k_y a\right)\cos\left(\frac{3}{2} k_x a\right)}
\end{aligned}$$



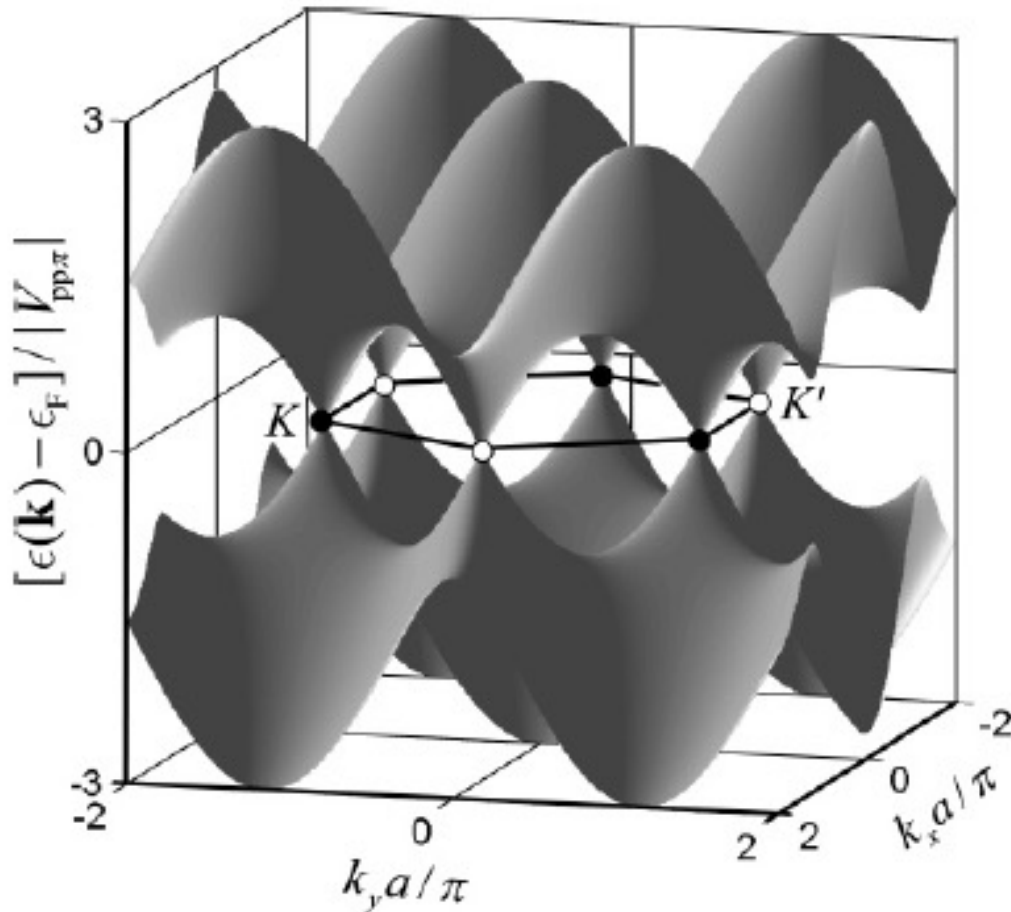
$$a = \sqrt{3}a_{C-C} = 2.49 \text{ \AA} \quad a_{C-C} = 1.44 \text{ \AA}$$

$$\vec{a}_1 = a \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) \quad \vec{a}_2 = a \left(\frac{\sqrt{3}}{2}, -\frac{1}{2} \right)$$

1st Brillouin Zone

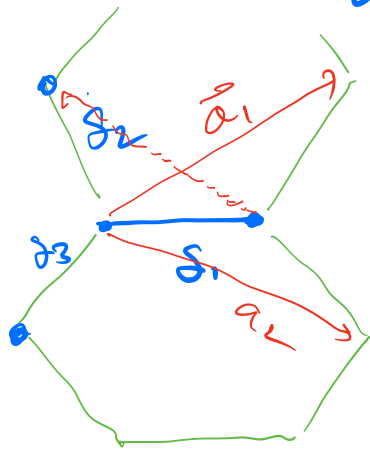


$$\varepsilon_k = \pm t \sqrt{1 + 4 \cos\left(\sqrt{3}k_x \frac{a}{2}\right) \cos\left(k_y \frac{a}{2}\right) + 4 \cos^2\left(k_y \frac{a}{2}\right)}$$



More elegant way to calculate $S(\vec{k})$

$$= \sum_i e^{i\vec{k} \cdot \vec{r}_i}$$



$$\vec{r}_1 = \vec{r}$$

$$\vec{r}_2 = \vec{r} - \vec{a}_2$$

$$\vec{r}_3 = \vec{r} - \vec{a}_1$$

$$S(\vec{k}) = e^{i\vec{k} \cdot \vec{r}} + e^{i\vec{k} \cdot (\vec{r} - \vec{a}_2)} + e^{i\vec{k} \cdot (\vec{r} - \vec{a}_1)}$$

$$= e^{i\vec{k} \cdot \vec{r}} [1 + e^{-i\vec{k} \cdot \vec{a}_2} + e^{-i\vec{k} \cdot \vec{a}_1}]$$

$$S^* S = \left[1 + e^{-i\vec{k} \cdot \vec{a}_1} + e^{-i\vec{k} \cdot \vec{a}_2} + e^{-i\vec{k} \cdot (\vec{a}_1 - \vec{a}_2)} + 1 + e^{i\vec{k} \cdot \vec{a}_1} + e^{i\vec{k} \cdot \vec{a}_2} + e^{i\vec{k} \cdot (\vec{a}_1 - \vec{a}_2)} + 1 \right]$$

$$= 3 + 2\cos(\vec{k} \cdot \vec{a}_1) + 2\cos(\vec{k} \cdot \vec{a}_2) + 2\cos(\vec{k} \cdot (\vec{a}_1 - \vec{a}_2))$$

check at Γ, M, K

$$\textcircled{1} \Gamma: \vec{k} = 0 \quad \cos(\) = 1$$

$$S^* S = 9$$

$$E = \pm |t| \sqrt{9} = \pm 3|t|$$

$$S = 3, \quad H = 3|t| \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Same Hamiltonian as diatomic molecule

for $\Sigma = +3t$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = + \begin{pmatrix} C_A \\ C_B \end{pmatrix} \quad C_B = C_A \quad \psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Sigma = -3t$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = - \begin{pmatrix} C_A \\ C_B \end{pmatrix} \quad \psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

M point $k = \frac{\vec{b}_1 + \vec{b}_2}{2}$

$$\vec{k} \cdot \vec{a}_1 = \pi \quad \vec{k} \cdot \vec{a}_2 = \pi \quad k \cdot (a_1 - a_2) = 0$$

$$\begin{aligned} S^\dagger S &= 3 + 2\cos\pi + 2\cos\pi + 2\cos 0 \\ &= 3 - 2 - 2 + 2 = 1 \end{aligned}$$

$$\Sigma = \pm |t|$$

K point $k = \frac{\vec{b}_1 - \vec{b}_2}{2}$

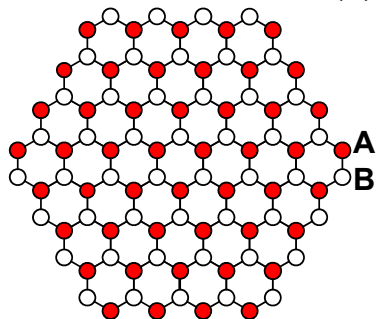
$$k \cdot a_1 = \frac{2\pi}{2} \quad k \cdot a_2 = -\frac{2\pi}{3} \quad k \cdot (a_1 - a_2) = \frac{4\pi}{3}$$

$$\begin{aligned} S^\dagger S &= 3 + 2\cos\left(\frac{2\pi}{2}\right) + 2\cos\left(-\frac{2\pi}{3}\right) + 2\cos\left(\frac{4\pi}{3}\right) \\ &= 0 \end{aligned}$$

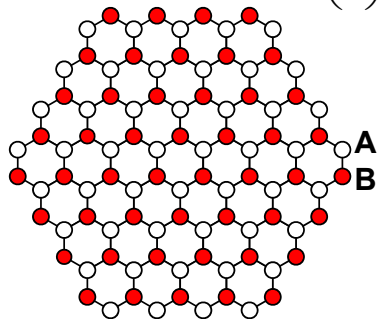
States at Γ point

Bloch states:

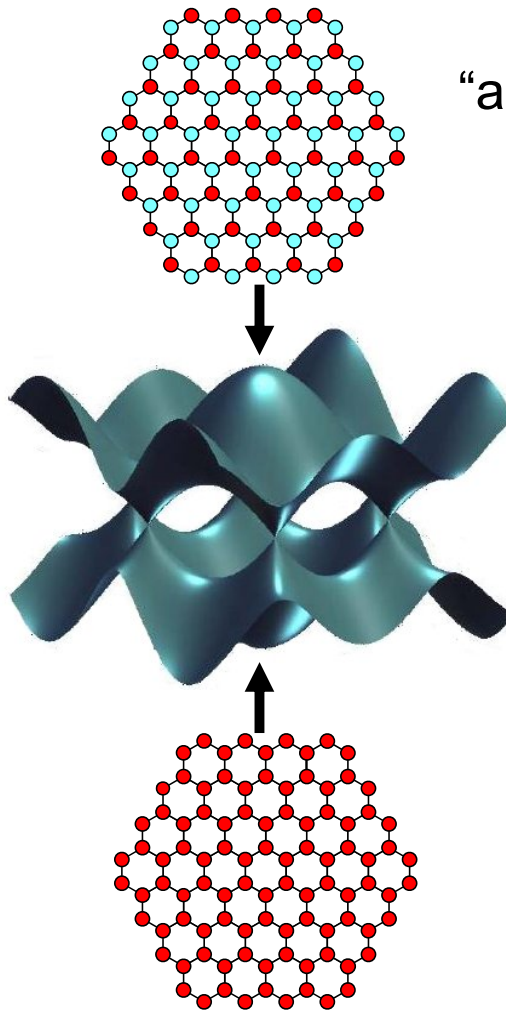
$$F_A(r), \quad \text{or} \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



$$F_B(r), \quad \text{or} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



Γ point:
 $k = 0$



“anti-bonding”

$$E = +3\gamma_0$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

“bonding”

$$E = -3\gamma_0$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Expansion around K point

$$K \oplus q \quad k = K + q$$

$$\text{use } K = \frac{b_2 - b_1}{3}$$

$$S = \underbrace{e^{ik \cdot \delta_1}} \left[1 + e^{-ik \cdot a_1} + e^{-ik \cdot a_2} \right]$$

$$K \cdot \delta_1 = \left(\frac{b_2 - b_1}{3} \right) \left(\frac{a_1 + a_2}{3} \right) = 0$$

$$e^{-ik \cdot a_1} = e^{-ik \cdot a_1} e^{-iq \cdot a_1}$$

$$K \cdot a_1 = -\frac{2}{3} \pi \quad \text{so } e^{-ik \cdot a_1} = \frac{1}{2} - i\frac{\sqrt{3}}{2}$$

$$\text{for small } q, \quad e^{-iq \cdot a_1} = 1 - iq \cdot a_1$$

$$\Rightarrow e^{-ik \cdot a_1} = \left(\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) (1 - iq \cdot a_1)$$

$$e^{-ik \cdot a_2} = \left(\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) (1 - iq \cdot a_2)$$

$$\text{add } \left[1 + e^{-ik \cdot a_1} + e^{-ik \cdot a_2} \right]$$

terms without q add to zero...

$$\begin{aligned}
 S &= -\frac{\sqrt{3}}{2} \mathbf{q} \cdot \mathbf{a}_1 + \frac{i}{2} \mathbf{q} \cdot \mathbf{a}_1 + \frac{\sqrt{3}}{2} \mathbf{q} \cdot \mathbf{a}_2 + i \mathbf{q} \cdot \mathbf{a}_2 \\
 &= -\frac{\sqrt{3}}{2} \mathbf{q} \cdot \underbrace{(\mathbf{a}_1 - \mathbf{a}_2)}_{\sqrt{3} a_{cc} \hat{y}} + \frac{i}{2} \mathbf{q} \cdot \underbrace{(\mathbf{a}_1 + \mathbf{a}_2)}_{\frac{3}{2} a_{cc} \hat{x}} \\
 &= \frac{3}{2} a_{cc} \cdot (i q_x - q_y) \rightarrow \frac{3}{2} a_{cc} (q_x + i q_y) \\
 &\quad \quad \quad \rightarrow x(i)
 \end{aligned}$$

$$H = \frac{3 a_{cc}}{2} t \cdot \begin{pmatrix} 0 & (q_x + i q_y) \\ (q_x - i q_y) & 0 \end{pmatrix}$$

solve $\det(H - \epsilon I) = 0$

$$\epsilon = \pm \frac{3}{2} a_{cc} \cdot |t| \cdot \sqrt{q_x^2 + q_y^2}$$

$$= \pm \frac{3}{2} a_{cc} \cdot |t| \cdot |q| \quad \text{linear bandstructure}$$

(Dirac cones)



note: at $k' = -k$, 1st term
is $(\vec{a}_2 - \vec{a}_1)$ so get

$$\mathcal{H} = \frac{3\alpha a t}{2} \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}$$

Gaps in graphene

We can open a gap by breaking
the symmetry between the A and B
sublattices

$$\varepsilon(A) - \varepsilon(B) = 2\Delta \quad \text{for some reason...}$$

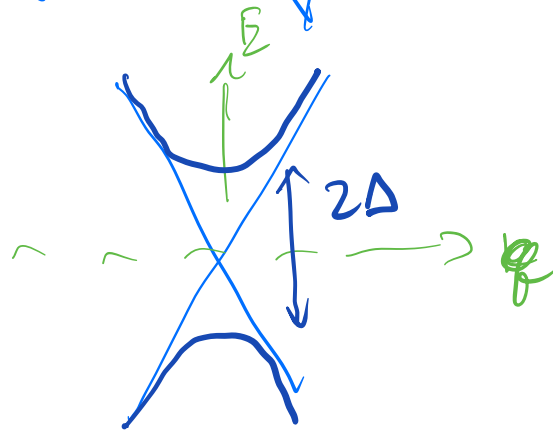
$$\mathcal{H} \Rightarrow \begin{pmatrix} \Delta & \frac{3\alpha a t}{2} (q_x - iq_y) \\ \frac{3\alpha a t}{2} (q_x + iq_y) & -\Delta \end{pmatrix}$$

↑
correction from
lecture

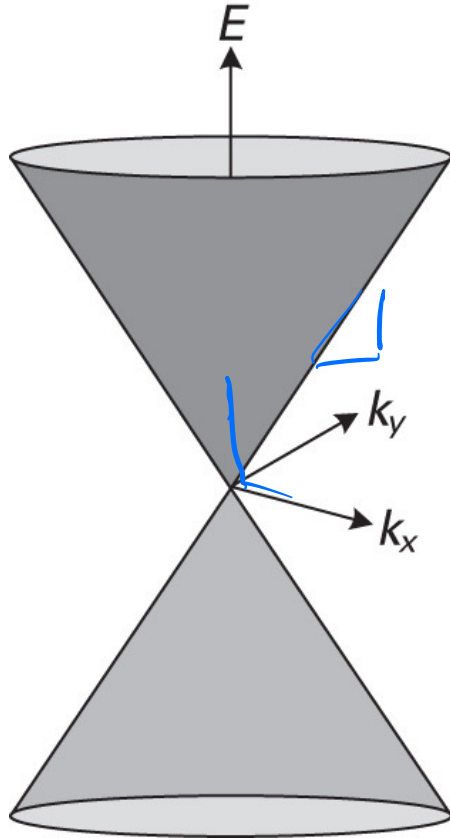
$$\det (H - \varepsilon I) = 0$$

$$\varepsilon^2 - \Delta^2 = \frac{3}{2} a_{\text{eff}} \hbar^2$$

$$\varepsilon(q) = \pm \sqrt{\frac{3}{2} a_{\text{eff}} \hbar^2 + \Delta^2}$$



Dispersion around K point



$$E = \pm \frac{3}{2} a c c' (t) |g|$$

switch back to k

$$v_F = \frac{1}{\hbar} \frac{dE}{dk} = \frac{3}{2} \frac{a c c' (t)}{\hbar}$$

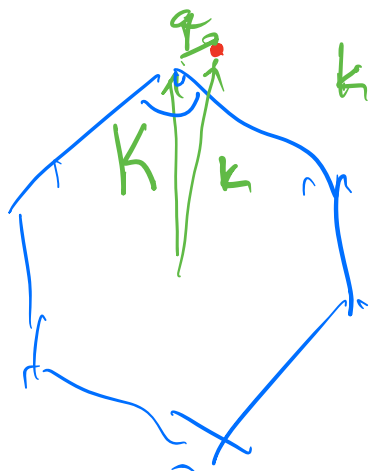
$$\sim 10^6 \text{ m/s}$$

$$(\frac{1}{300} c)$$

$$\hbar v_F = 6.3 \text{ eV} \cdot \text{\AA}$$

$$\frac{\text{eV}}{\hbar} = 1.5 \times 10^{15} \text{ s}^{-1}$$

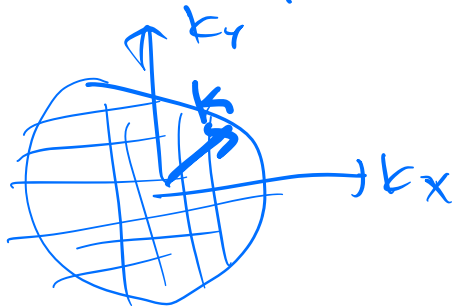
Wong Fig.3.9



$$k = K + q$$

↓
call this k

Density of States



$$N(k) = \frac{\pi k^2}{4\pi^2 / \text{Area}}$$

$\times 4$ ← 2 valley
 $\times 2$ ← 2 spin

$$n(k) = \frac{N}{\text{Area}} = \frac{k^2}{\pi}$$

$$k_F = \sqrt{\pi n}$$

plug in numbers: $n = 10^{12} \text{ cm}^{-2}$

$$k_F = 1.8 \times 10^6 \text{ cm}^{-1} = 0.018 \text{ \AA}^{-1}$$

$$k_F = \frac{4\pi}{3a} \approx 1.8 \text{ \AA}^{-1} \quad k_F \approx 1\% \text{ of } k_{\text{max}}$$

$$E_F = (6.3 \text{ eV} \cdot \text{\AA}) \cdot (0.018 \text{ \AA}^{-1}) \approx 0.1 \text{ eV}$$

$$E_F = v_F k_F = v_F \sqrt{\pi n}$$

States near the K point

* switch to be consistent with Kaneelson

$$\mathcal{H}_K = \frac{3a_{cc}|t|}{2} \begin{pmatrix} 0 & q_x^{-i}q_y \\ q_x^{+i}q_y & 0 \end{pmatrix} \quad \mathcal{H}_K' = \propto \begin{pmatrix} 0 & q_x^{+i}q_y \\ q_x^{-i}q_y & 0 \end{pmatrix}$$

$$\Sigma = \pm \frac{3}{2} \frac{a_{cc}}{2} |t| \cdot |q|$$

$$\begin{pmatrix} 0 & q_x^{-i}q_y \\ q_x^{+i}q_y & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = \pm |q| \begin{pmatrix} C_A \\ C_B \end{pmatrix}$$

$$\begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0 \end{pmatrix} \begin{pmatrix} C_A \\ C_B \end{pmatrix} = \pm \begin{pmatrix} C_A \\ C_B \end{pmatrix}$$

$$+ \quad e^{-i\phi} C_B = C_A \Rightarrow C_B = e^{i\phi} C_A$$

$$- \quad e^{-i\phi} C_B = -C_A \Rightarrow C_A = -e^{-i\phi} C_B$$

so we can write states as:

$$\psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix} \quad \psi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} -e^{i\phi} \\ 1 \end{pmatrix}$$

or $e^{-i\phi/2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\phi/2} \\ e^{i\phi/2} \end{pmatrix}$

and $-e^{-i\phi/2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\phi/2} \\ -e^{i\phi/2} \end{pmatrix}$

* phase between A/B sublaters
is "pseudospin"

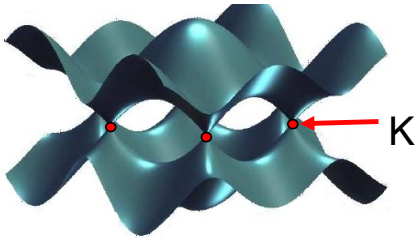
+ solution at $\phi = 0$ $\psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

states are $\Rightarrow$ $\phi = \pi$ $\psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
orthogonal

Spinor, not vector!

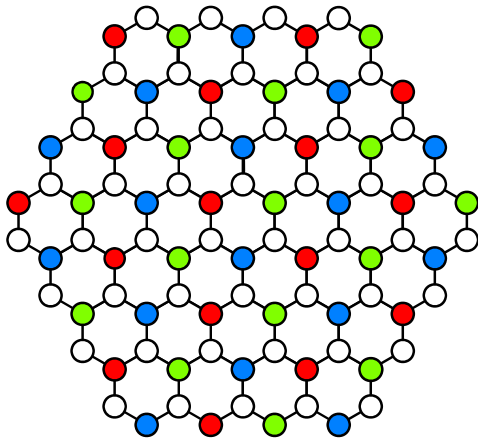
$\langle \psi_1 | \psi_2 \rangle = 0$ when 180° apart

States at K point

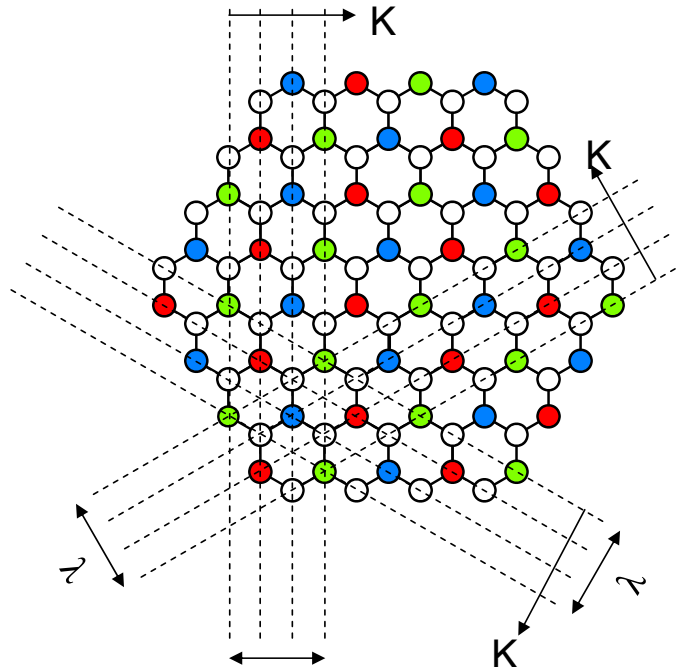


$$|\mathbf{K}| = \frac{4\pi}{3a} \Rightarrow \lambda = \frac{3a}{2}$$

$$F_A(r), \quad \text{or} \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



$$F_B(r), \quad \text{or} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



Phase:

- 1
- $e^{i\frac{2\pi}{3}}$
- $e^{i\frac{4\pi}{3}}$